

CS 3513 Numerical Methods

Programming Exercise 7

Due: 2025-04-25

last possible late submission : 2025-05-05

No initial code for this exercise. You may answer each question in a separate python file or place them all in one script. Place your code in the GitHub Classroom assignment repository. See canvas assignment page for repository invitation. Submit your solution by committing and pushing your code to your GitHub Classroom assignment repository. If you use jupyter, be sure to pair your jupyter notebook with the python file. We will use the python file for grading.

Question 1 Estimate $\int_0^1 e^{-(x^2)} dx$ using the following methods. Remember to print out the

question part label with the estimate, such as

```
print(" Part (a) integral is expected to be {:.16.14f}".format(expectedValue)).
```

Or if a part has multiple lines, start with a header line,

```
print(" Part (c)")
```

and then indent the rest of the output, such as,

```
print("      T(0) = {:.16.14f} with error {:.8.3e}:".format(T[0], abs(T[0]-expectValue))).
```

- (a) (2 points) From probability and statistics, this is the error function without the normalization term. The error function is available in python's `math.erf` function. We would expect the value of the integral to be `erf(1)*sqrt(pi)/2`. Print out this value. This value will let you know if your other estimates are in the ballpark of being correct.
- (b) (5 points) Plot the function $f(x) = e^{-(x^2)}$ over the range 0 to 1 and save the plot as "ex7q1plot.pdf". Look at the figure to verify that the function is well behaved over the interval.
- (c) (8 points) Using the iterative trapezoid equation (from page 7 of slides), calculate the first 8 estimates of the integral and the absolute value of the difference between the estimate and the expected value from part (a). That is, calculate $T(0)$, $T(1)$, $T(2)$, ..., $T(7)$, print each value with its error on a separate line.
- (d) (8 points) Using the trapezoid estimates from the previous part, calculate the Romberg integration of the integral keeping all the values of the table. Print out the values of the diagonal of the table and the absolute value of the difference between that value and the expected value from part (a), each on its own line.
- (e) (3 points) Use `scipy's integrate.quad` to estimate the integral. Print out the quad estimate and the absolute value of the difference between that value and the expected value from part (a).
- (f) (2 points) Print out the number of function evaluations that `scipy's quad` used and the number of function evaluations that Romberg integration used.

Question 2 Estimate $\int_{1.5}^4 \left(\frac{200}{2x^3 - x^2} \right) \left(5 \sin \frac{20}{x} \right)^2 dx$ using the following methods. Remember to print out the question part label with the estimate. Or if a part has multiple lines, start with a header line, and then indent the rest of the output.

- (a) (5 points) Plot the function $f(x) = e^{-(x^2)}$ over the range 0 to 1 and save the plot as "ex7q2plot.pdf". Look at the figure to verify that the function is well behaved over the interval and note that it has a lot more curvature than the function from question 1.
- (b) (3 points) Not wanting to use calculus to work through solving the definite integral for this problem, we will start with scipy's adaptive quadrature function, `integrate.quad`, to make the integral estimate. Print out quad's estimate, quad's estimated absolute error, and the number of function evaluations taken.
- (c) (8 points) Using the iterative trapezoid equation (from page 7 of slides), calculate the first 8 estimates of the integral. For each estimate, print the value of the estimate and the absolute value of the difference between the trapezoid estimate and quad estimate from part (b).
- (d) (8 points) Using the trapezoid estimates from the previous part, calculate the Romberg's integration of the integral keeping all the values of the table. Print out the values of the diagonal of the table and the absolute value of the difference between that value and the quad's estimate from part (b), each on its own line. Note that Romberg's estimates don't start improving until the step size of the underlying trapezoid estimate is small enough to capture the basic form of the function.

Question 3 We have the following table of volume (in cubic inches) and pressure (in pounds per square inch) measurements taken of a gas as it expands.

Volume, v	0.75	1.00	1.25	1.50	1.75	2.00	2.25	2.50	2.75
Pressure, p	89.8	68.7	55.0	45.8	39.3	34.4	30.5	27.5	26.0

We would like to estimate the work, W , done by the gas using the relation

$$W = \int_{0.75}^{2.75} p \, dv.$$

- (5 points) First, Plot the table with v as the x-axis values, p as the y-axis values. Use a dashed line connection between data points to reinforce the idea that we don't know the underlying function. Also use large red dots at each data point. To do this with matplotlib, specify the extra settings using keyword parameters to the plot call, such as `linestyle='--'`, `marker='o'`, `markersize=12`, `markerfacecolor='red'`, `markeredgecolor='black'`, `linewidth=2`, `color='blue'`. Save the plot as "ex7q3plot.pdf". Look at the figure to verify that the data looks is well behaved.
- (2 points) Integrate the data using scipy's `integrate.trapezoid` function. Print out the result. You can leave the result in inch-pounds instead of converting it to the more common foot-pound.
- (2 points) Integrate the data using scipy's `integrate.simpson` function. Print out the result. You can leave the result in inch-pounds instead of converting it to the more common foot-pound.
- (3 points) Integrate the data using scipy's `integrate.romb` function. Use the parameter `show=True` to print the Romberg table.
- (3 points) Which provides a better estimate, Romberg or Simpson? Why?